

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	20 Jan 2026
Team ID	LTVIP2026TMIDS24102
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

Technical Architecture: The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

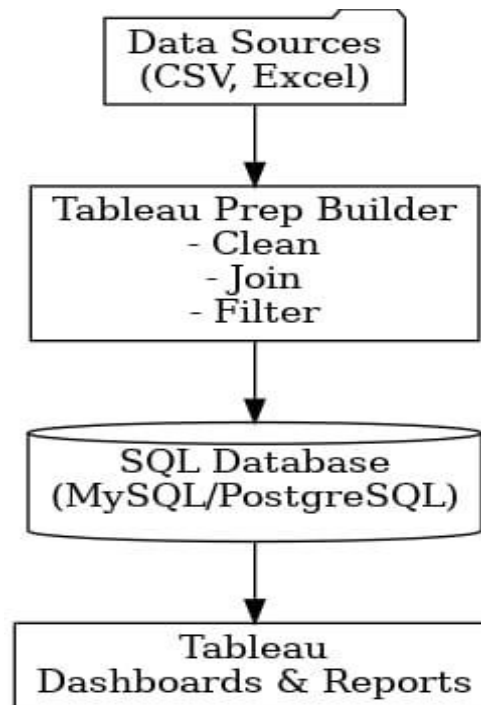


Table-1: Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Users interact with dashboards, filters, maps, charts, forecasting visuals	Tableau Public (Web UI), HTML (embedded dashboard if hosted)
2	Application Logic-1	Data cleaning, transformation, filtering, calculated fields	Tableau Calculations, Tableau Prep
3	Application Logic-2	Forecasting logic and trend analysis	Tableau Forecasting (Exponential Smoothing Model)
4	Application Logic-3	Aggregations, grouping by region/category, KPI generation	Tableau Aggregations (SUM, COUNTD, AVG), Parameters
5	Database	Structured dataset containing UNESCO heritage site details	CSV Dataset (UNESCO_cleaned.csv)
6	Cloud Database	Cloud-based hosting of dashboard and dataset	Tableau Public Cloud
7	File Storage	Storage of raw and cleaned dataset files	Local File System (.csv files)
8	External API-1	Map rendering and geographic visualization	Tableau Mapbox Integration
9	External API-2	OpenStreetMap geographical data service	OpenStreetMap (via Tableau)
10	Machine Learning Model	Time-series forecasting of heritage site trends	Tableau Built-in Forecast Model
11	Infrastructure (Server / Cloud)	Deployment and hosting of dashboards	Tableau Public Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open geographic map services used in dashboards	OpenStreetMap
2	Security Implementations	Access control handled by Tableau Public platform	Tableau Public Access Control
3	Scalable Architecture	Separation of Data Layer, Processing Layer, Visualization Layer (3-Layer Architecture)	Tableau 3-Tier Architecture
4	Availability	Cloud-hosted dashboards accessible 24/7 via web	Tableau Public Cloud Hosting

5	Performance	Optimized using aggregated calculations and filtered dataset	Tableau Extract Optimization
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